QA

Welcome to ASDXDDL8M3- Data Modelling and Management

Moving around

Navigation

Right click + drag / Space bar + drag = Panning mode
+ / Mouse wheel = Zoom in
- / Mouse wheel = Zoom out
Ctrl + 2 = Zoom to selected object
Ctrl + 1 = 100% zoom
Ctrl + 0 = Adjust zoom to show all content
↑,↓,←,→ = Move on the board
Ctrl + Shift + [1-9,a-z] = Jump to quick link

Shortcuts to use

Selection

Click = Single object selection
Shift / CTRL + click = Multi selection
Shift + click & drag = Area selection
R = Toggle area selection
Ctrl + A = Select all
Alt + ↑,↓,←,→ = Select next object in the direction
Alt + Shift + ↓,↑ = Select group child or parent
Esc = Deselect object and reset to standard selection

Tools

N = Add sticky note
G = Add rectangular sticky note
S = Add shape
T = Create text
C = Add comment
L =Add line
D = Draw with nib brush
P = Previous brush in drawing mode
E = Eraser in drawing mode
Ctrl + L = Lock/unlock object
Ctrl + Shift + P = Pin/unpin object
Ctrl + H = Hide/unhide object

Have a play

Use the tools to get familiar with adding content.

Team 1

Add some post its!

Write something

Team 2

Add some post its!

Write something

Team 3

Add some post its!

Write something

Team 4

Add some post its!

Write something

Housekeeping

Rules regarding webcams

Keep your Webcam on throughout the delivery

Rules regarding mute

Keep your Microphone off throughout for privacy unless you are asked or need to respond.

Frequency of breaks

Short Breaks are scheduled every hour to support you with your learning.

Rules regarding group participation

Everyone should equally participate in Group discission and take lead in contributing your view.

Rules regarding questions

The Golden Rule:
'There is no such thing as a stupid question'
'Even when asked by an instructor'
Please have a go at answering questions
Also:
'A question never resides in a single mind'
By asking questions, you are helping everyone out!

Anything else

Teams

♞	Ayotunde	Jonathan	Robyn					
💎	Chris	Sharon	Tracy					
🪐	Andrew	Helen	Rachel					
⚡	Charlie	Ryan	Josh					

♞ 💎 🪐 ⚡

Activity 1: Movies Database Normalisation

Learner instructions

We have been given the task of creating an online resource based on movies produced in the last 50 years.

One of the features of this resource will be a RDMS that will store details of the personnel and production information relating to a large number of films.

We have an Excel file provided with data that has been partly normalised in that some of the entities and their attributes have already been identified.

However one of the sheets on the Excel file labelled Movies and Actors still needs to be normalised before it is suitable for the RDMS.

Screenshots can be seen below of this table and its attributes.

Task 1

The attributes in the Movies and Actors table are:

Director
Actor
gender
character_name
cast_order
movie_id
title
budget
homepage
overview
popularity
release_date
revenue
runtime
movie_status
tagline
vote_average
vote_count

Identify and name potential entities from the information supplied.

Remember that there may be many- to- many relationships between the tables that would require a bridging table (a table containing the foreign keys of the tables to be joined)

Think about the fact that and actor can play many characters but these characters can appear in a number of movies i.e. Luke Skywalker appears in Star Wars but also in The Empire Strikes Back!

Also when you create any of the main tables from they will need a Primary Key attribute that may need to be added for example director_ID

Identify and name potential entities from the information supplied.

Hint:

There are at least 5 entities needed and 2 of these are bridging tables.

1. Director
2. .....
3. .....
4. .....
5. .....

Task 2

Compile data dictionaries for your identified entities.
You may use the pro forma on the following page (or any other valid format).
You should include any Keys (Primary and Foreign), Data Types for each attribute and any suitable constraints I.e. Null, Not Null

Entity description			
Database Entity name			
Key			
	Attribute	Data Type	Constraints/comments

Task 3

Identify entity relationships. Draw an **Entity Relationship Diagram** showing the relationships (cardinality) between the entities(tables) you identified in Task 1.

Show the Primary and Foreign keys to show how tables would relate to each other i.e. The Primary key of one table (Parent) would become the Foreign key in its related table (Child)

Hint:

Bridging tables that are used to avoid many- to- many relationships are basically formed of two one- to- many relationships.

The diagram below shows the "many- to- many" relationship to avoid duplicate values or multiple values in the one column, between Orders and Products. OrderID and ProductID have become foreign keys in the OrderDetails table.

Replace the image above with your own Entity Relationship Diagram (ERD)

https://online.visual- paradigm.com/

Movies and Actors Database

The attributes in the Movies and Actors table are:
Director
Actor
gender
character_name
cast_order
movie_id
title
budget
homepage
overview
popularity
release_date
revenue
runtime
movie_status
tagline
vote_average
vote_count

Task 1
Identify and name potential entities from the information supplied.

Identify and name potential entities from the information supplied.
Hint:
There are at least 5 entities needed and 2 of these are bridging tables.

1. Director
2. Movie
3. Actor
4. Actor - Movie - Character
5. Director - Movie

Task 2
Compile data dictionaries for your identified entities.
You may use the pro forma on the following page (or any other valid format).
You should include any Keys (Primary and Foreign), Data Types for each attribute and any suitable constraints I.e. Null, Not Null

Entity Description	Stores information about directors		
Database Name	Movie Database		
Entity Name	Director		
Key	Attribute	Data Type	Constrains (Null/Not Null)
PK	Director_ID	INT	NOT NULL
	Director	VARCHAR(150)	NOT NULL

Entity 2

Entity Description	Stores information about each movie including release details and ratings.		
Database Name	Movie Database		
Entity Name	Movie		
Key	Attribute	Data Type	Constrains (Null /Not Null)
PK	Movie_ID	INT	NOT NULL
	Title	Varchar (150)	NOT NULL
	Popularity	Decimal (15,6)	NOT NULL
	Release_Date	Date	NOT NULL
	Runtime	INT	NOT NULL
	Vote_average	Decimal(5,2)	NOT NULL
		MONEY	NULL
	Homepage	VARCHAR(MAX)	NULL allowed
	Populairty	FLOAT(53)	NULL allowed
	Revenue	MONEY	NULL allowed
	Movie_Status	VARCHAR(MAX)	NULL allowed
		VARCHAR(150)	NULL allowed
	Tagline	VARCHAR(100)	NULL allowed
	Vote_Count	IN-	

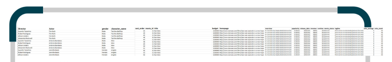
Entity 3

Entity Description	Stores information about characters portrayed by actors in movies		
Database Name	Movie Database		
Entity Name	Actor		
Key	Attribute	Data Type	Constrains (Null/Not Null)
PK	Actor_ID	INT	NOT NULL, PRIMARY KEY, auto generated
	actor_Name	VARCHAR(255)	NOT NULL
	gender	VARCHAR(30)	NULL, controlled list recommended

Entity 4

Entity Description	Bridge Linking Directors and Movies		
Database Name	Movie Database		
Entity Name	Movie_Director		
Key	Attribute	Data Type	Constrains (
FK	Movie_id	INT	Movie Refere
FK	director_id	INT	Director F direc

Movies and Actors Database



The attributes in the Movies and Actors table are:
Director
Actor
gender
character_name
cast_order
movie_id
title
budget
homepage
overview
popularity
release_date
revenue
runtime
movie_status
tagline
vote_average
vote_count

Task 1
Identify and name potential entities from the information supplied.

Identify and name potential entities from the information supplied.
Hint:
There are at least 5 entities needed and 2 of these are bridging tables.

1.Director
2.Actor
3.Movie
4.Actor-Movie-Character
5.Director-Movie

Task 2
Compile data dictionaries for your identified entities.
You may use the pro forma on the following page (or any other valid format).
You should include any Keys (Primary and Foreign), Data Types for each attribute and any suitable constraints I.e. Null, Not Null

Entity Description	Director info		
Database Name	MovieDB		
Entity Name	Director		
Key	Attribute	Data Type	Constrains (Null/Not Null)
Primary Key	DirectorID	Int	Not null
	Director	nVarchar,50	Not null

Entity 2

Entity Description	Movie information		
Database Name	MovieDB		
Entity Name	Movie		
Key	Attribute	Data Type	Constrains (Null/Not Null)
Primary Key	MovieID	Int	Not Null
Foreign Key	Director_Id	iN	Not null
	title	nVarChar, 50	Not Null
	release_date	Date	Not Null
	budget	Money	Not Null
	revenue	Money	Not Null
	runtime	Int	Not Null
	movie_status	nVarChar, 50	Not Null
	tagline	nVarChar, MAX	Null
	overview	nVarChar, MAX	Not Null
	homepage	nVarChar, 50	Null
	popularity	Float 9	Not Null
	vote_average	Decimal 2,1	Not Null
	vote_count	Int	Not Null

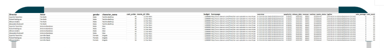
Entity 3

Entity Description	Actor Information		
Database Name	MovieDB		
Entity Name	Actor		
Key	Attribute	Data Type	Constrains (Null/Not Null)
Primary key	Actor_id	Int	Not Null
	actor_name	nVarchar,50	Not Null
	Gender	Varchar	Not Null

Entity 4

Entity Description	Actor-Movie-Character		
Database Name	MovieDB		
Entity Name	Actor-Movie-Character		
Key	Attribute	Data Type	Constrains (
FK	Actor_ID	INT	No
FK	Movie_ID	INT	No
	cast_order	INT	No
	Character_name	nVarchar,50	No

Movies and Actors Database



The attributes in the Movies and Actors table are:
Director
Actor
gender
character_name
cast_order
movie_id
title
budget
homepage
overview
popularity
release_date
revenue
runtime
movie_status
tagline
vote_average
vote_count

Task 1
Identify and name potential entities from the information supplied.

Identify and name potential entities from the information supplied.
Hint:
There are at least 5 entities needed and 2 of these are bridging tables.

1.Director
2.Actor
3.Movie
4.Actor Character Movie
5.Director Movie

Task 2
Compile data dictionaries for your identified entities.
You may use the pro forma on the following page (or any other valid format).
You should include any Keys (Primary and Foreign), Data Types for each attribute and any suitable constraints I.e. Null, Not Null

Entity Description	Director Info		
Database Name	movies DB		
Entity Name	Director		
Key	Attribute	Data Type	Constrains (Null/Not Null)
Primary Key	Director ID	Integer	Not Null
	Director	Varchar (50)	Not Null

Entity 2

Entity Description	Actor Information		
Database Name	MovieDB		
Entity Name	Actor		
Key	Attribute	Data Type	Constrains (Null/Not Null)
Primary Key	ActorID	Integer	Not null
	Actor	Varchar(50)	Not null
	Gender	Varchar(20)	Null allowed

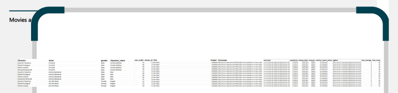
Entity 3

Entity Description	Movie Information		
Database Name	movieDB		
Entity Name	Movie		
Key	Attribute	Data Type	Constrains (Null/Not Null)
Primary Key	MovieID	Interger	Not Null
	Title	Varchar (50)	Not Null
	Budget	Money	Null Allowed
	Homepage	Varchar (225)	Null Allowed
	Overview	Text	Null Allowed
	Popularity	Decimal(10,2)	Null Allowed
	ReleaseDate	Date	Not Null
	Revenue	Money	Null Allowed
	Runtime	Integer	Null Allowed
	MovieStaus	Varchar(50)	Not Null
	Tagline	Varchar(50)	Null Allowed
	VoteAverage	Decimal(3,1)	Null Allowed
	VoteCount	Integer	Null Allowed

Entity 4

Entity Description	Actor appearance in move information	
Database Name	movieDB	
Entity Name	Actor Character Movie	
Key	Attribute	Data Type
Foreign Key	ActorID	Integer
Foreign Key	MovieID	Integer
	CharacterName	Varchar(50)
	CastOrder	Integer

Movies a



The attributes in the Movies and Actors table are:
Director
Actor
gender
character_name
cast_order
movie_id
title
budget
homepage
overview
popularity
release_date
revenue
runtime
movie_status
tagline
vote_average
vote_count

Task 1
Identify and name potential entities from the information supplied.

Identify and name potential entities from the information supplied.
Hint:
There are at least 5 entities needed and 2 of these are bridging tables.

1. Director
2. Movie
3. Actor
4. Character-Actor-Movie
5. Director-Movie

Task 2
Compile data dictionaries for your identified entities.
You may use the pro forma on the following page (or any other valid format).
You should include any Keys (Primary and Foreign), Data Types for each attribute and any suitable constraints I.e. Null, Not Null

Entity Description	Actor Info		
Database Name	MovieDB		
Entity Name	actor		
Key	Attribute	Data Type	Constrains (Null/Not Null)
Primary Key	actor_ID	Int	NOT NULL
	actor	nVarchar(50)	NOT NULL
	gender	nVarchar(15)	NOT NULL

Entity 2

Entity Description	Director Info		
Database Name	MovieDB		
Entity Name	director		
Key	Attribute	Data Type	Constrains (Null/Not Null)
Primary Key	director_ID	Int	NOT NULL
	director_name	nVarchar(50)	NOT NULL

Entity 3

Entity Description	Movie Info		
Database Name	Movie DB		
Entity Name	movie		
Key	Attribute	Data Type	Constrains (Null/Not Null)
Primary Key	movie_ID	Int	NOT NULL
	title	nVarchar(100)	NOT NULL
	budget	Money	NOT NULL
	homepage	nVarchar(150)	NULL
	overview	nVarchar(Max)	NULL
	popularity	Float(53)	NULL
	release_date	Date	NOT NULL
	revenue	Money	NOT NULL
	runtime	Int	NOT NULL
	movie_status	nVarchar(20)	NOT NULL
	tagline	nVarchar(Max)	NULL
	vote_avg	Decimal(2,1)	NOT NULL
	vote_count	Int	NOT NULL

Entity Description	Director-Movie		
Database Name	MovieDB		
Entity Name	Director-Movie-bridge		
Key	Attribute	Data Type	Constrains (Nu
Foreign Key	movie_ID	Int	NOT N
Foreign Key	director_ID	Int	NOT N